

PAH-OMC-028: Normal contractions on the minimal closure

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claim C6-SPACETIME-SIGNATURE

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Purpose and scope

This is the one gate-level synthesis for T-095 and R-567. It repairs the nonlinear closure argument of R-530 on the exact R-512 target. The full certificate and preregistration are in `strategy/pa-hyp/`, under `PAH-OMC-028-certificate.md` and `PAH-OMC-028-prereg-v1.json`. No microscopic definition is changed. The state is the R-510 labelled all- Q probability law. The original two-row strip, $K = 2$, $M_s = 1$, $\epsilon = 1/2$, $\beta = \nu = 1$, $m^2 = \theta = 0$, and other OMC-016 couplings equal to one are inherited. Its j -before-anchored- n construction is not repeated or exchanged.

The statement under inherited hypotheses

Let H be the real invariant state- L^2 space, D the exact bounded gauge-invariant, globally amplitude- ℓ^1 -Lipschitz finite-prefix cylinders, and

$$E(f, g) = \frac{1}{2} \sum_r \nu[c_r \Delta_r f \Delta_r g].$$

R-511 supplies the dense domain, the nonnegative conductances and this identity. R-512 supplies the minimal closure $(\overline{E}, V)$, with $\|u\|_V^2 = \|u\|_H^2 + \overline{E}(u, u)$ and an injective continuous inclusion into H . These are inherited analytic hypotheses.

For every scalar η with $\eta(0) = 0$ and $|\eta(a) - \eta(b)| \leq |a - b|$, and every $u \in V$,

$$\eta(u) \in V, \quad \overline{E}(\eta(u), \eta(u)) \leq \overline{E}(u, u). \quad (1)$$

The existing spectral semigroup $T_{\min}(t)$ is then positivity preserving, L^∞ contractive and satisfies $T_{\min}(t)1 = 1$ for $t \geq 0$.

The precise flaw and its exact diagnostic

The R-530 closure paragraph passes from $E(\eta f) \leq E(f)$ directly to form-Cauchy behavior of $\eta(f_k)$. The implied difference estimate $E(\eta f - \eta g) \leq E(f - g)$ is false. To test that universal inequality, use probability weights $(1/2, 1/2)$, both directed rates two, and the directed-half convention. Then $E(h) = (h_0 - h_1)^2$. With $f = (1, 2)$, $g = (0, 1)$, and $\eta(x) = (0 \vee x) \wedge 1$,

$$E(f - g) = 0, \quad E(\eta f - \eta g) = 1, \quad \|f - g\|_V^2 = 1, \quad \|\eta f - \eta g\|_V^2 = \frac{3}{2}.$$

Each single-vector inequality remains true. This diagnostic rejects the paired inequality; it does not refute (1), establish a non-Cauchy image sequence, or substitute a finite model for PAH. The original R-530 bytes are preserved, with this proof superseding only the flawed inference.

Core stability

Scalar composition preserves finite-prefix dependence and gauge invariance. If $|f| \leq M$, then $|\eta f| \leq M$, and its amplitude Lipschitz constant does not increase. Thus $\eta(D) \subset D$. The scalar Lipschitz bound integrates to

$$\|\eta f - \eta g\|_H \leq \|f - g\|_H.$$

This also proves quotient well-definedness and continuity on H . On D , the rootwise squared bound, weighted by the unchanged nonnegative conductances, proves $E(\eta f) \leq E(f)$.

Primary proof: closed energy epigraph

Put $q(u) = \overline{E}(u, u)$ for $u \in V$. The set

$$C = \{(u, a) \in H \times \mathbb{R} : u \in V, q(u) \leq a\}$$

is closed. Indeed, for $(u_k, a_k) \rightarrow (u, a)$ in $H \times \mathbb{R}$, the energies and H norms are bounded. Choose an energy-liminf subsequence; Hilbert weak compactness supplies a weakly V -convergent further subsequence. Its weak H limit is the strong H limit u , so $u \in V$. Weak lower semicontinuity gives $q(u) \leq \liminf q(u_k) \leq a$. This lower semicontinuity follows because the energy balls are convex and norm-closed in V , hence weakly closed by separation. Domain membership is part of the argument.

Given any $u \in V$, take $f_k \in D$ with $f_k \rightarrow u$ in V . The energy seminorm triangle inequality gives $E(f_k) \rightarrow q(u)$. Core contraction gives $(\eta f_k, E(f_k)) \in C$, while H continuity gives $\eta f_k \rightarrow \eta u$. Closedness of C proves (1). No form-Cauchy hypothesis on ηf_k is inserted. Growing supports are allowed, and no root formula on all of V is assumed.

Independent proof: convex averages

The same $w_k = \eta f_k$ is bounded in V . A weakly convergent subsequence has limit $v = \eta u$ by strong H convergence. Mazur's lemma yields finite convex combinations z_m of successive tails with $\|z_m - v\|_V < 1/m$. They belong to D . Convexity implies

$$E(z_m) \leq \sup_{\ell \geq m} E(w_{k_\ell}) \leq \sup_{\ell \geq m} E(f_{k_\ell}) \rightarrow q(u).$$

Their strong V limit gives $q(v) \leq q(u)$. The convex combinations are approximants in the proof, with no averaging of rates or states.

Semigroup and conservation

For the self-adjoint $K \geq 0$ associated with q , the resolvent $J_\lambda = (I + \lambda K)^{-1}$ uniquely minimizes $q(v) + \|v - h\|_H^2/\lambda$. If $0 \leq h \leq 1$, clipping to $[0, 1]$ decreases both terms. Uniqueness forces $0 \leq J_\lambda h \leq 1$. The spectral limit $(J_{t/m})^m \rightarrow e^{-tK}$ in H preserves that closed order interval. Positivity extends by bounded positive truncation.

The original D contains 1, the state is a probability, and all root differences of 1 vanish. Thus $q(1) = 0$ and form Cauchy-Schwarz gives $\overline{E}(1, v) = 0$ for every $v \in V$. The associated-operator definition implies $1 \in \text{Dom } K$ and $K1 = 0$. Consequently $e^{-tK}1 = 1$. Positivity and linearity now give the L^∞ contraction. Preservation of $[0, 1]$ alone would not prove conservation.

Literature and proof coverage

Fukushima, Oshima and Takeda, *Dirichlet Forms and Symmetric Markov Processes*, second edition (2011), Section 1.1, Theorem 1.4.1 and Theorem 1.4.2(v), supply the standard context. Only the forward form-to-semigroup implication is used; no locally compact topology or Hunt process is imported. This is a PAH proof repair using standard functional analysis, not a new general Dirichlet-form theorem.

Five Lean theorems verify the paired energy and form-norm counterexamples, the arbitrary single-root inequality, the arbitrary-sequence closed epigraph passage, and the scalar zero-energy cross-term lemma. Epigraph closedness from the Hilbert completion, Mazur’s lemma, the inherited measure construction and spectral representation are analytic here. Exact rational primary and independent generator-matrix checks agree. External review of the analytic crosswalk and closure proof is invited.

Devil’s-advocate

1. Single-vector contraction implies pairwise energy contraction: UPHELD objection; the exact counterexample rejects this shortcut.
2. Positivity alone implies closedness: UPHELD objection. For point evaluation energy $q(f) = f(0)^2$, let $s_n(x) = (1 - nx)_+$ on $[0, 1]$. Then $1 - s_n \rightarrow 1$ in L^2 , but $q(1 - s_n) = 0 < q(1) = 1$. The spikes alone diagnose nonclosability; their complements exhibit the lower-semicontinuity failure. R-512 closedness is essential.
3. Negative conductances or a two-Lipschitz map preserve the inequality: DISMISSED by explicit hostile controls; both violate named hypotheses.
4. Bounded energy proves domain membership without closedness: UPHELD objection; weak compactness in V and injectivity provide it here.
5. Sub-Markov implies conservation: UPHELD objection; a killed scalar form has resolvent $J_1 1 = 1/2$. Zero energy of the actual constant is used.
6. Target Markov structure implies finite-to-target convergence: UPHELD objection; the PAH-OMC-020 identification gate remains open.

Result footer

Result ID: R-567; contract PAH-OMC-028; task T-095
Claim ID: C6-SPACETIME-SIGNATURE
Precise statement: Every scalar normal contraction preserves the R-512 minimal form domain and decreases its energy. The associated semigroup preserves positivity, bounded norm and constant one.
Scope: Full minimal domain; inherited R-510/R-511/R-512 hypotheses.
Dependencies: Fixed probability state, exact core and closed form.
Evidence grade: ANALYTIC, EXACT, EXECUTED; partial Lean formalization.
Reproduction command: `python -X utf8 verification/scripts/pah_omc028_verify.py --check`
Expected output: PAH-OMC-028 PASS; five Lean theorems compile.
Falsification gate: Failure of state/core/closedness hypotheses; valid normal contraction with domain or energy violation.
Tier before / after: T1 / T1; auxiliary_support; T-054 unchanged.
No-overclaim statement: No OMC-020 convergence, Pre-A, spacetime, QFT, gravity, continuum, mass gap or TOE conclusion.
Next required action: Separately contract Markov-extension uniqueness for the same exact operator and state; no new model or owner scan.